

Continuous-time Controllers

Control Engineering

Course Schedule

1. Systems
2. Continuous-time Controllers
3. Laplace Domain
4. Poles/Zeros & Stability
5. Controller Design
6. Discrete-time Controllers
7. State Space Representation
8. Advanced Control Methods

Continuous-time Controllers

- Control objectives
- On-off controllers
- PID controllers
- Working point
- Proportional band

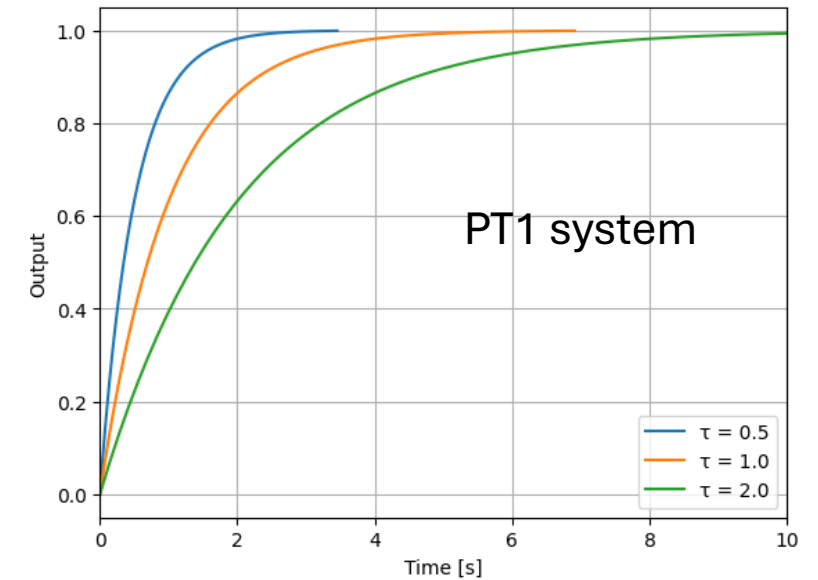
Control

Models (differential equations) of dynamic systems allow to calculate the effect of inputs on outputs

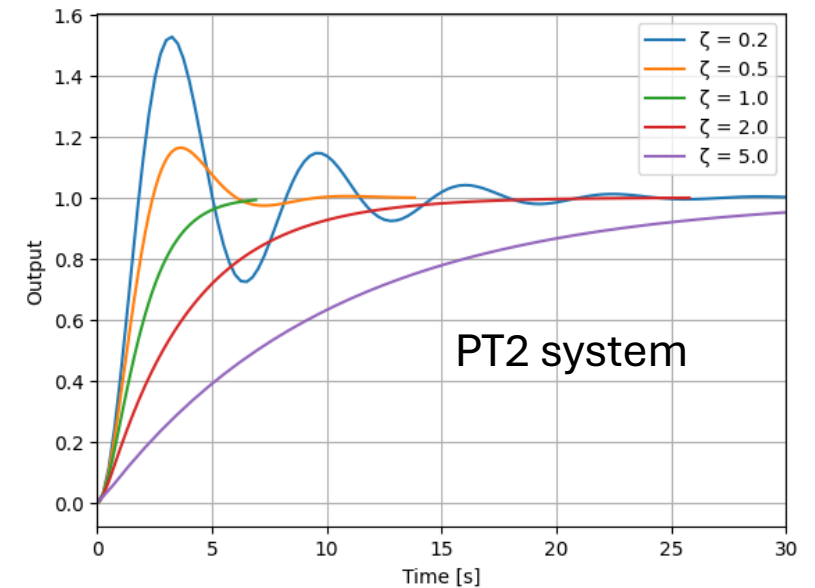
Idea of control: modify the input in order to obtain a desired output

Example: adjust step input to reach a desired steady state ($u_{\text{step}} = \frac{y(\infty)}{K}$)

Problems: response may be too slow, disturbances degrade performance

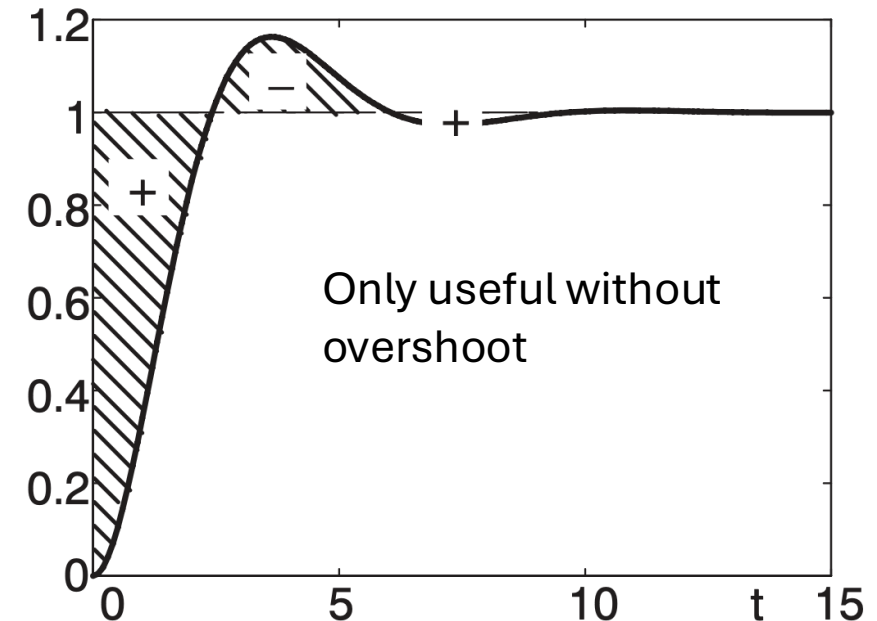
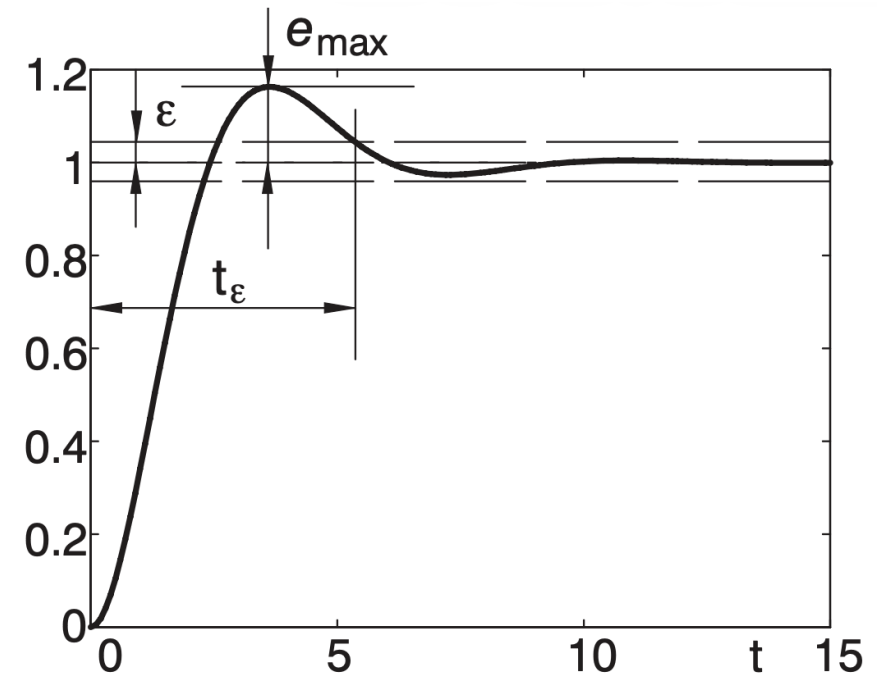


Step responses



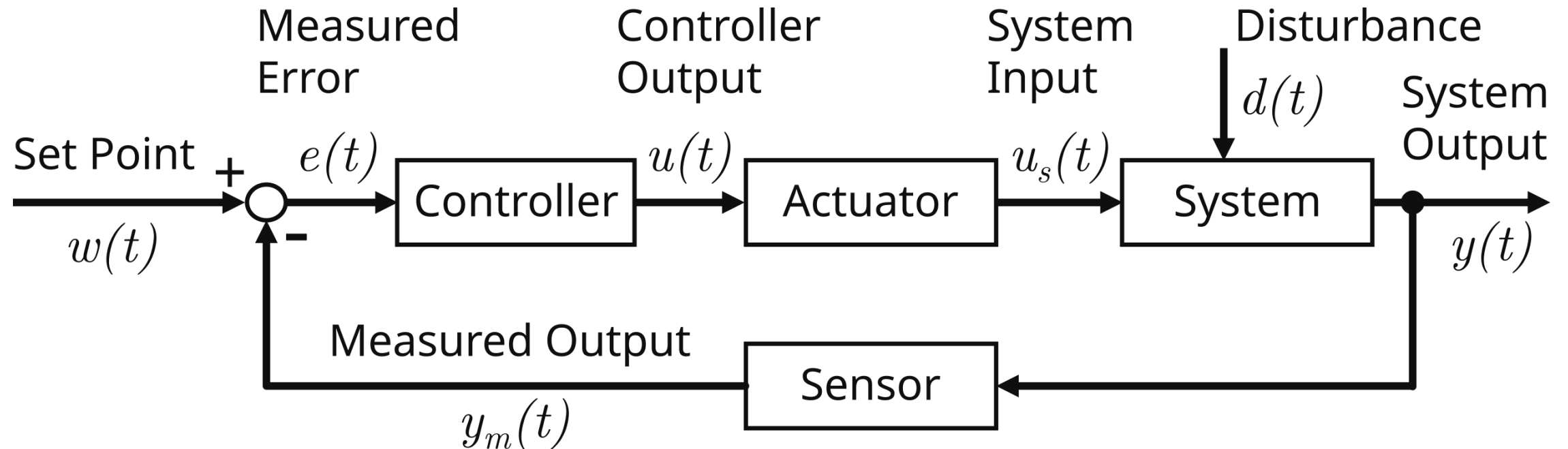
Control Objectives

- Stability (bounded output)
- Fast response: rise and settling time
- Low overshoot
- Small steady-state error
- Robustness (insensitivity to model uncertainty and disturbances)



Control Loop

- Feedback: measure system output, compare it to desired behavior (e.g., steady state), and use the resulting error as input to a controller
- Goal: reduce error



What Is a Controller?

System itself (subsystem of total control system), implemented in hardware or software

- Input: measured error
- Output: process input (low-energy control signal converted into physical action by actuator)

Hardware controllers

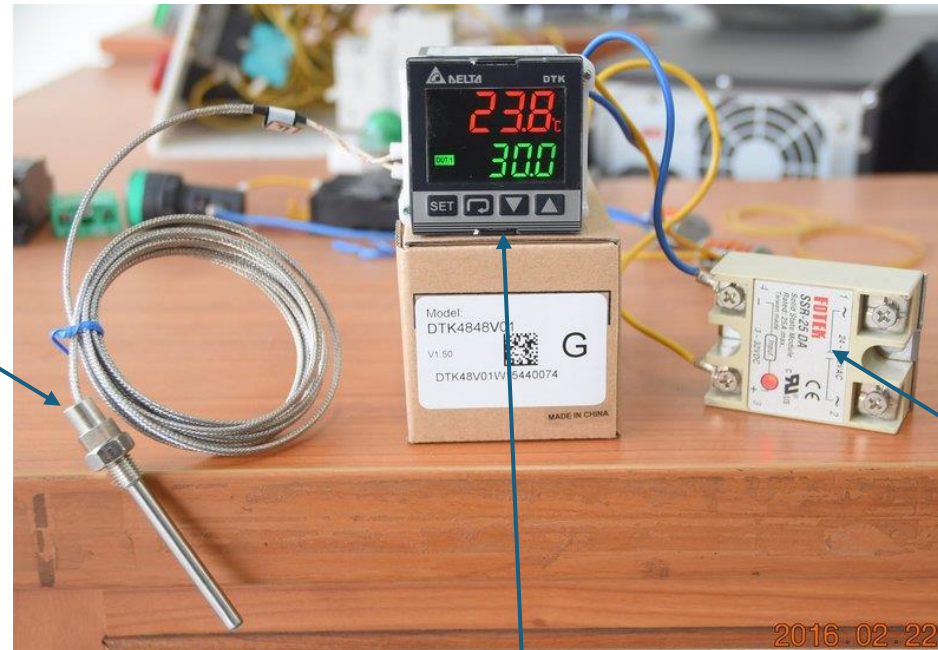
- Fast, deterministic, limited flexibility
- Examples: mechanical, analog electronic controllers using op-amps, FPGA-based (digital)

Software controllers (use of analog actuators via a digital-to-analog converter)

- Flexible, complex algorithms, sampling delays (dead time)
- Examples: Microcontroller-based, PLCs

Example: Temperature Control

Sensor:
resistance thermometer
to measure temperature
(system output)



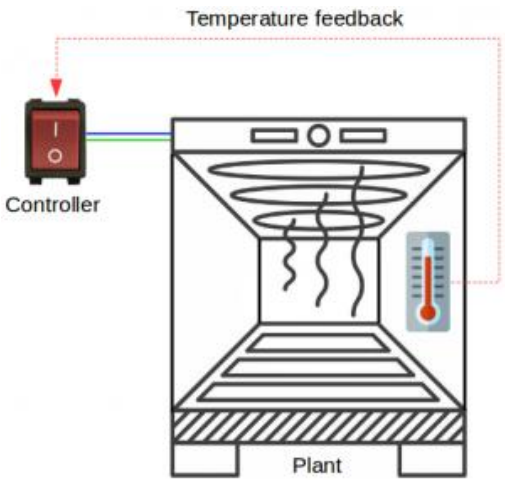
Actuator:
solid-state relay to
switch a heater/cooler
(plant) on or off

Controller:
digital electronic PID switching controller
to activate the relay (controller output)

On-Off Controller

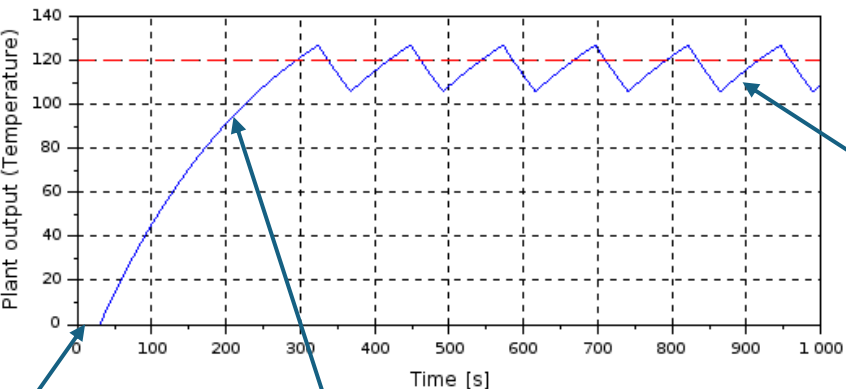
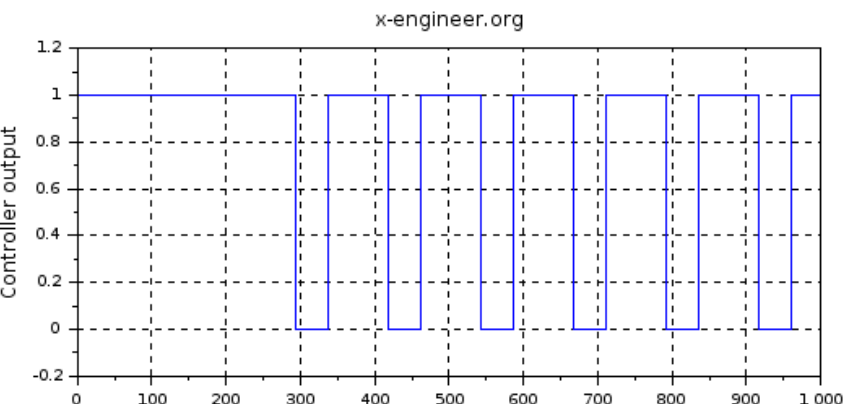
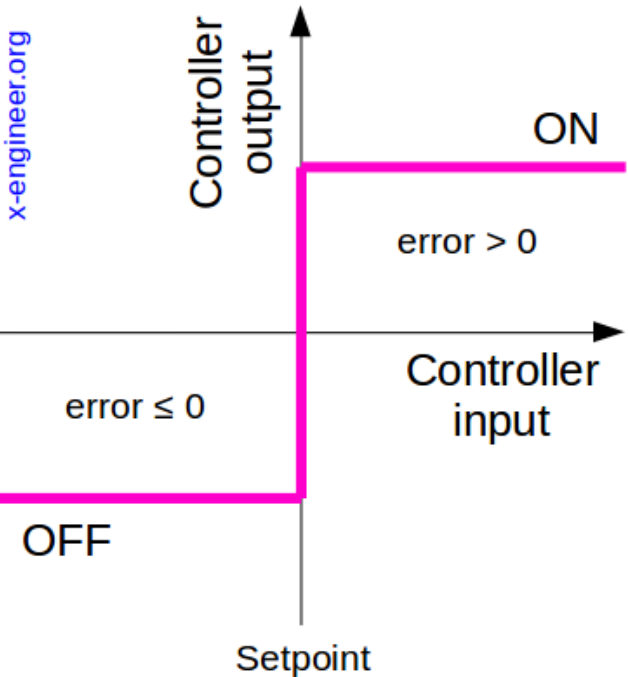
aka bang-bang control

Example: industrial oven



Issue:
High-frequency switching
stresses actuator

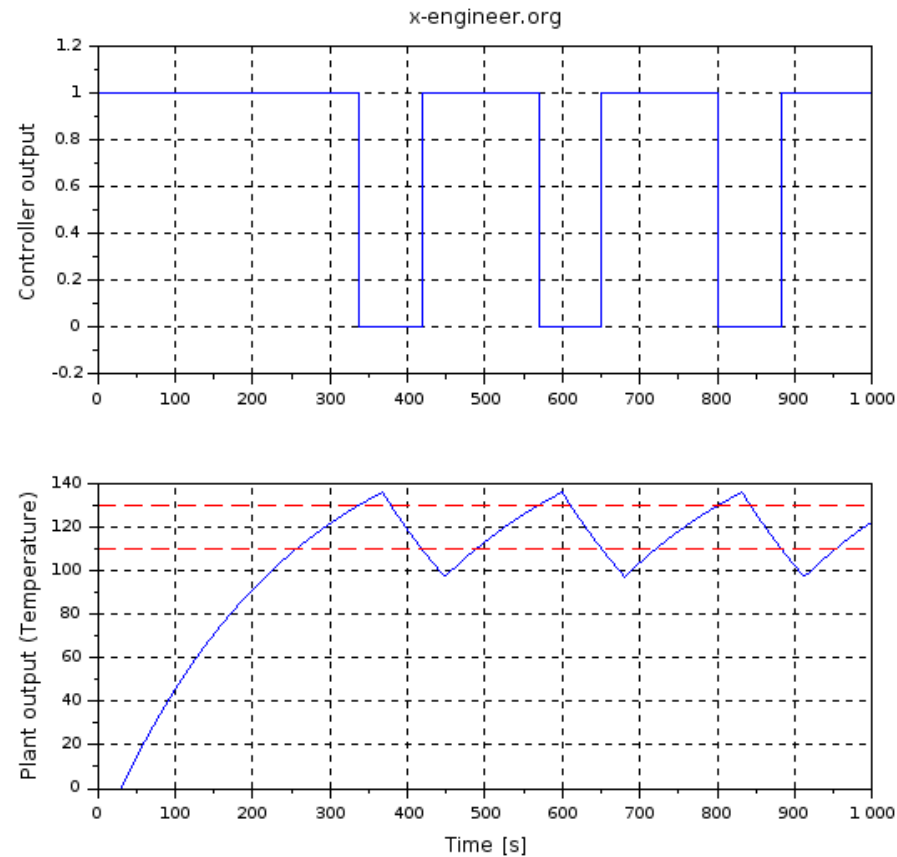
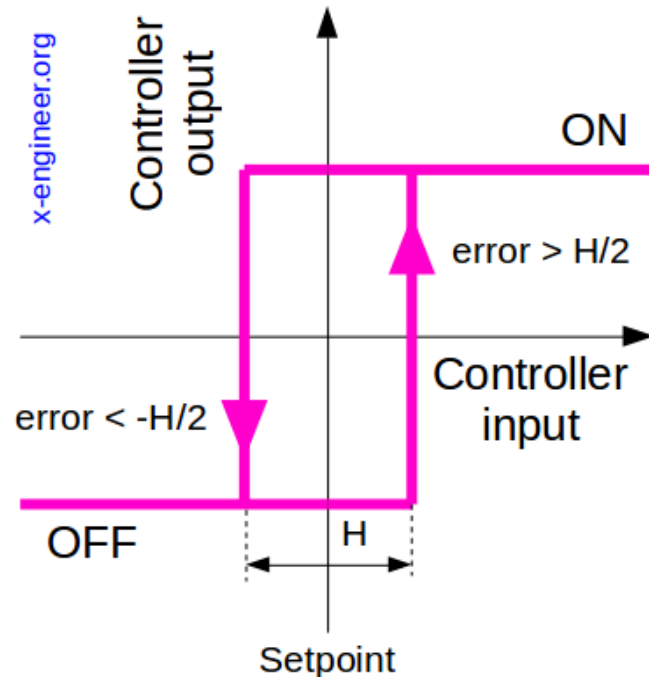
oscillations around setpoint



dead time

first-order process

On-Off Controller with Hysteresis



Less switching at the expense of accuracy

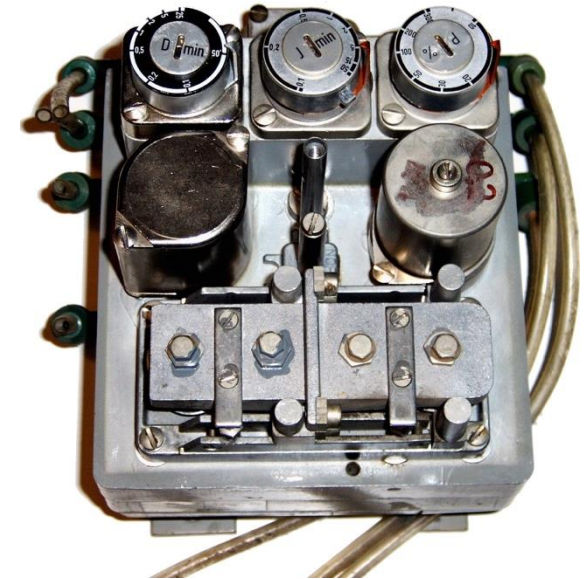
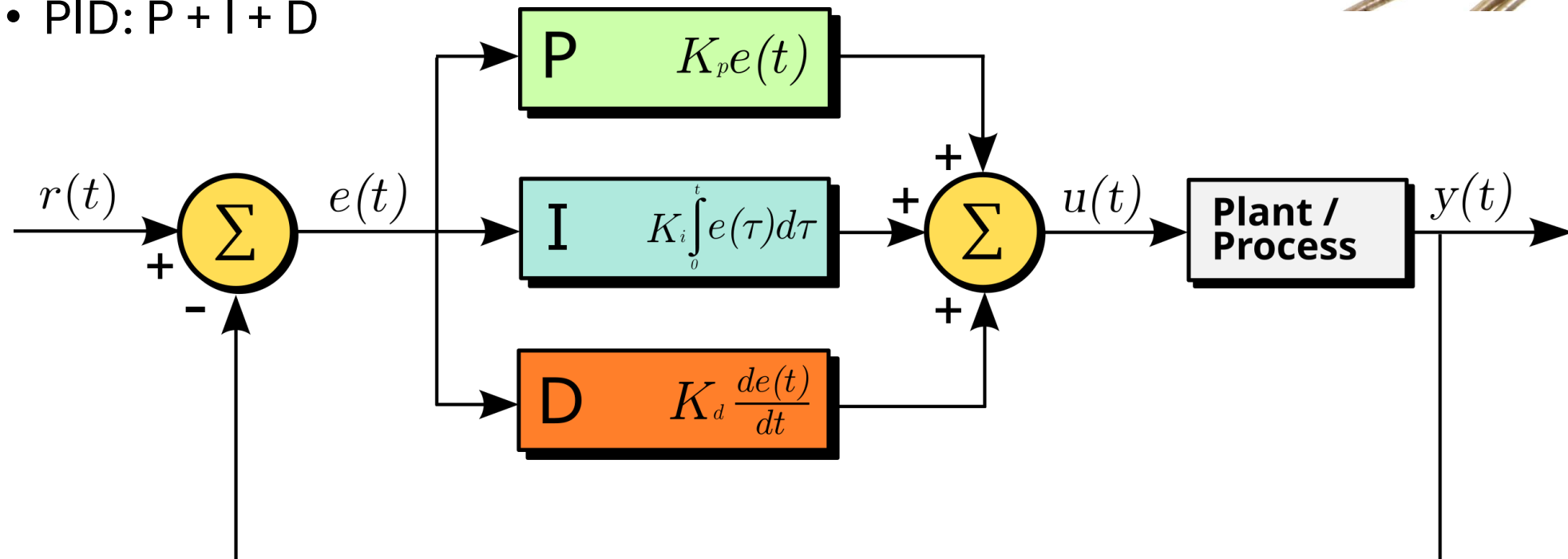
On-off controllers are simple and low-cost, but have limited precision.
→ Continuous controllers (e.g., PID)

PID Controllers

Most widely used industrial controller

- P
- PI: P + I
- PD: P + D
- PID: P + I + D

Parallel structure most common in practice



Proportional Control (P)

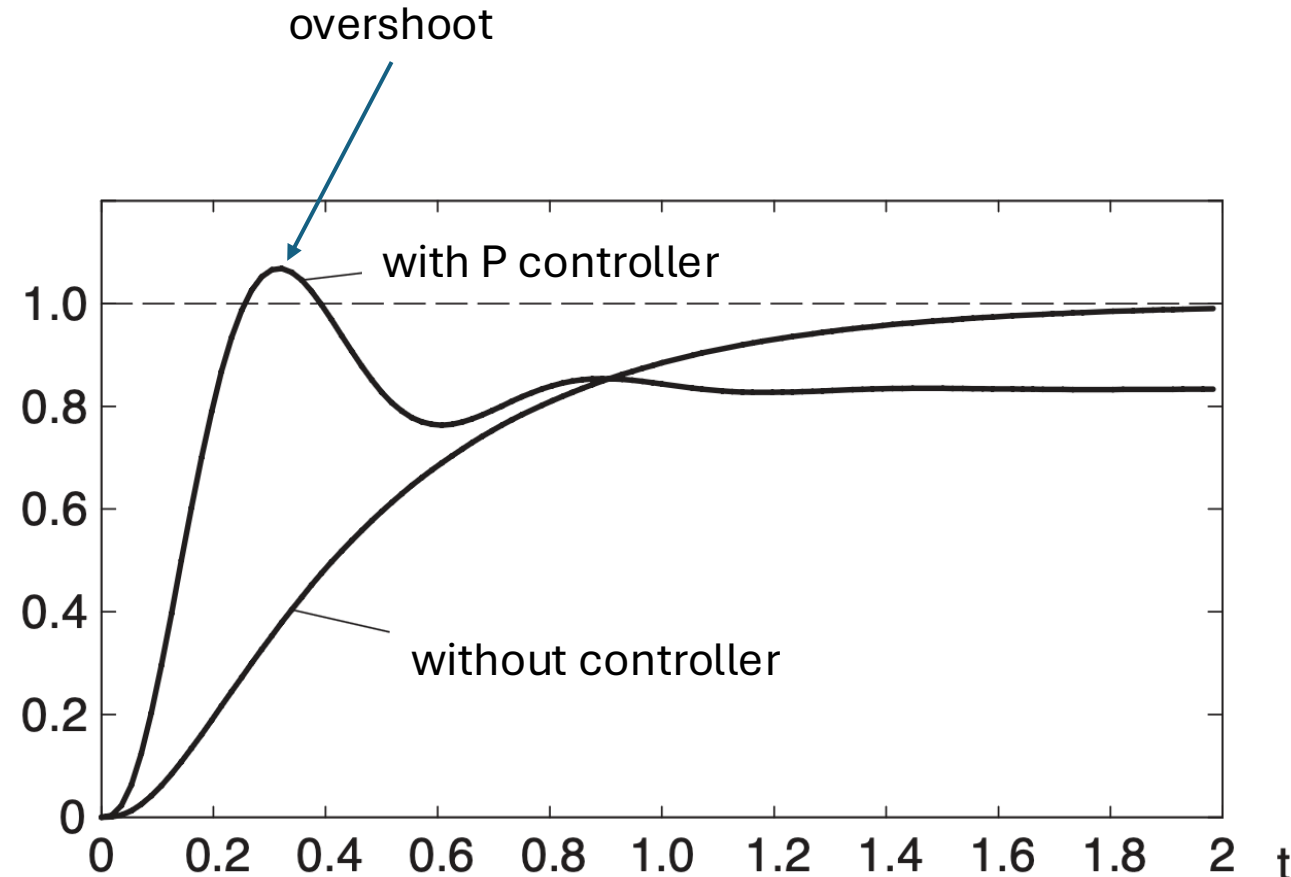
Control input proportional to error (static system):

$$u(t) = K_p \cdot e(t)$$

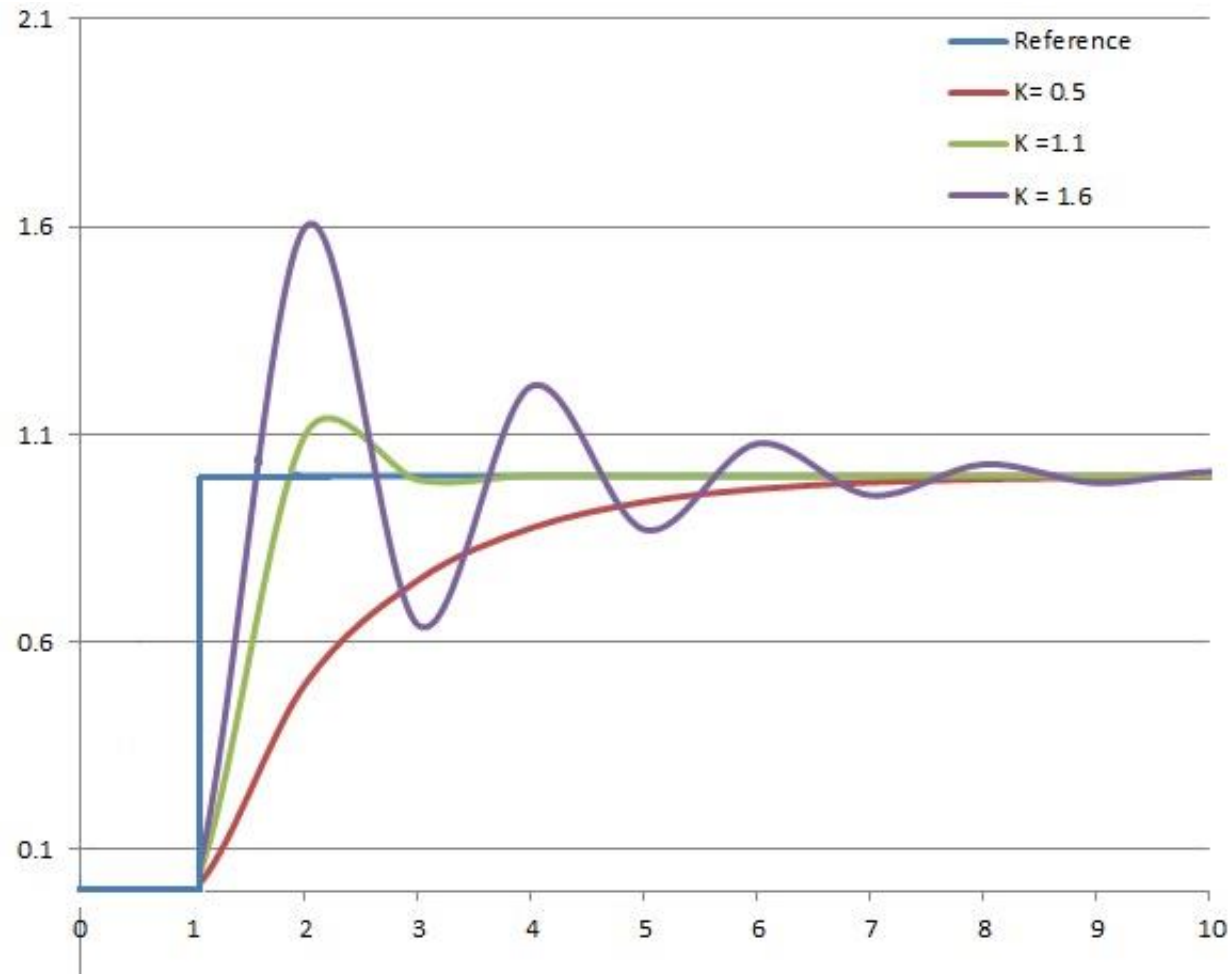
- Increases loop gain $\rightarrow$ faster response
- Reduces damping $\rightarrow$ more oscillation
- Steady-state error in many systems:

$$e(t) = 0 \Rightarrow u(t) = 0$$

$\rightarrow y(t)$ settles at some error, just big enough to produce $u(t)$ balancing the system dynamics (e.g., holding a mass up)



Effect of Proportional Gain Parameter



- With simple proportional control, oscillations only occur in second-order systems (or higher)
- Destabilization of a first-order system with proportional feedback can only occur with positive feedback

Historical: Centrifugal Governor in a Windmill

Working principle:

wind turns the blades

→ blades turn a shaft (plant)

→ shaft spins a pair of weighted arms (controller)

→ arms mechanically twist the blades (actuator)

Automatic regulation:

faster rotation (controlled variable: angular speed)

→ centrifugal force pushes weights outward (error: deviation from position at reference angular speed)

→ blades are twisted to catch less wind (proportional control) → slower rotation

Steady-state error: need for twisted blades to balance torque from wind and load → speed offset



Integral Control (I)

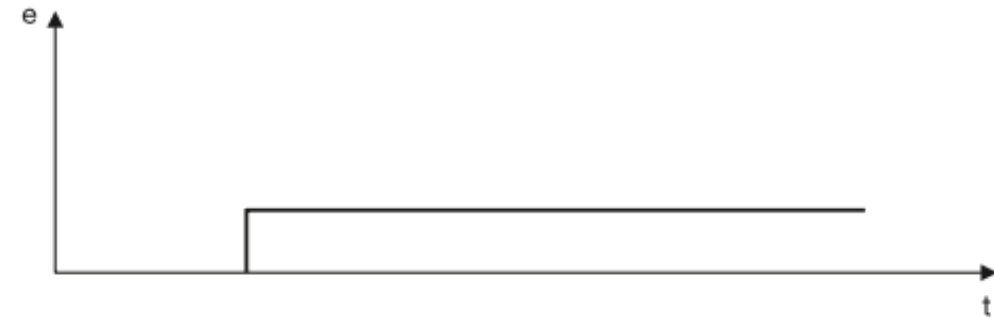
Integrator accumulating past error (dynamic)

- Eliminates steady-state error to step inputs
- Slower response
- Reduces damping → more oscillation

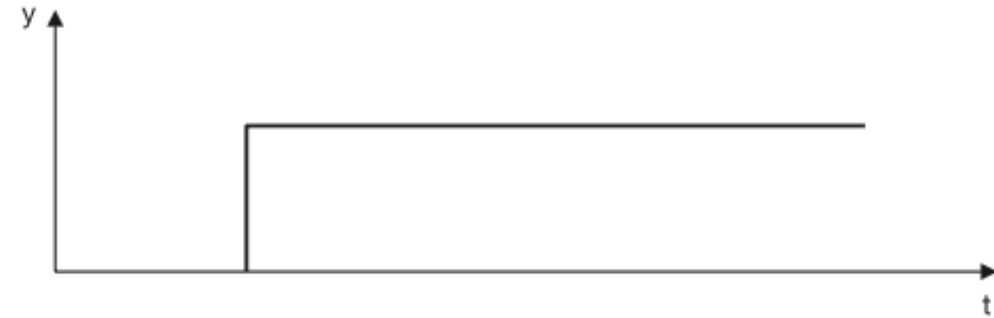
Order of total system increases (adds memory)

- First-order → second-order
- Second-order → third-order

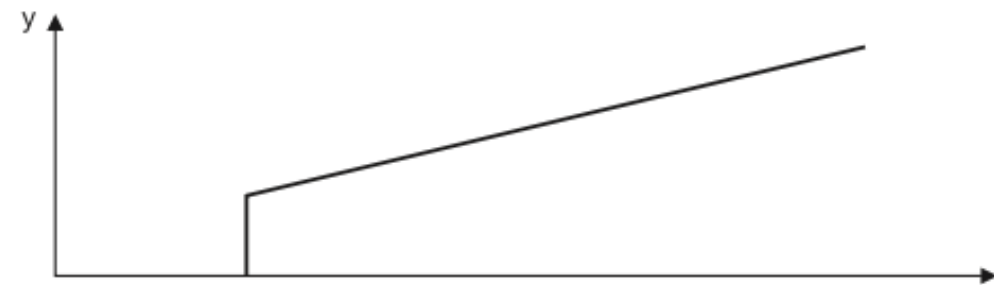
Step change of control deviation



Step responses
P controller



PI controller



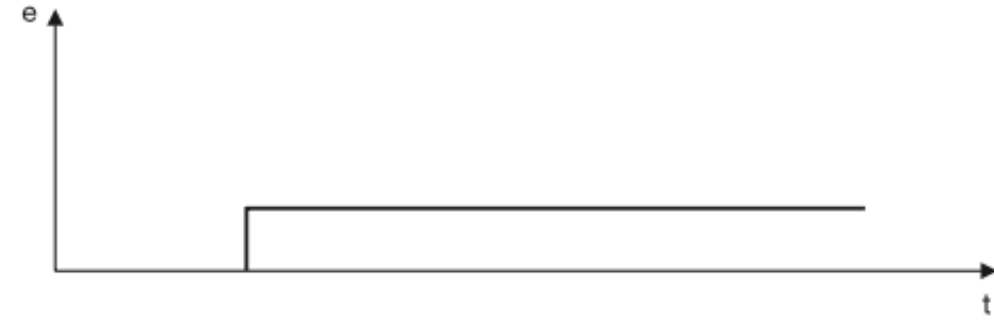
Derivative Control (D)

Derivative predicting future error based on the rate of change (dynamic)

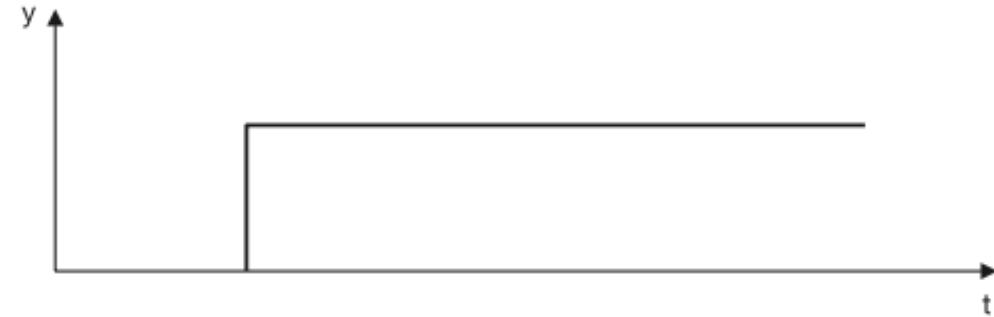
- Predictive action
- Increases effective damping (keeps system near critical damping): less overshoot, fewer oscillations)
- Derivative action amplifies high-frequency noise → often paired with a low-pass filter

Used less often in practice due to impact on system stability → PI

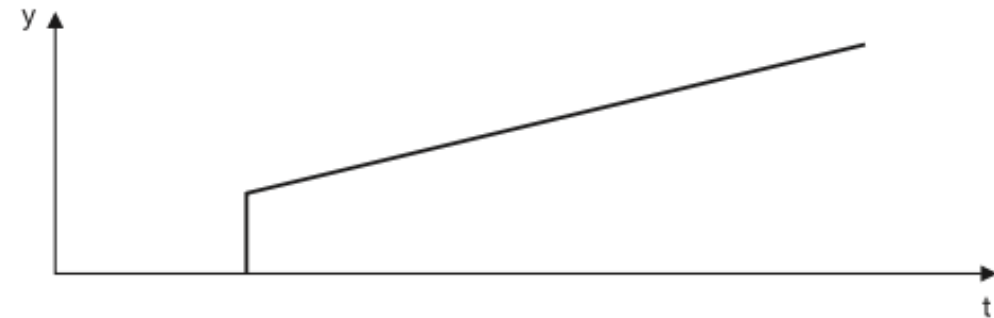
Step change of control deviation



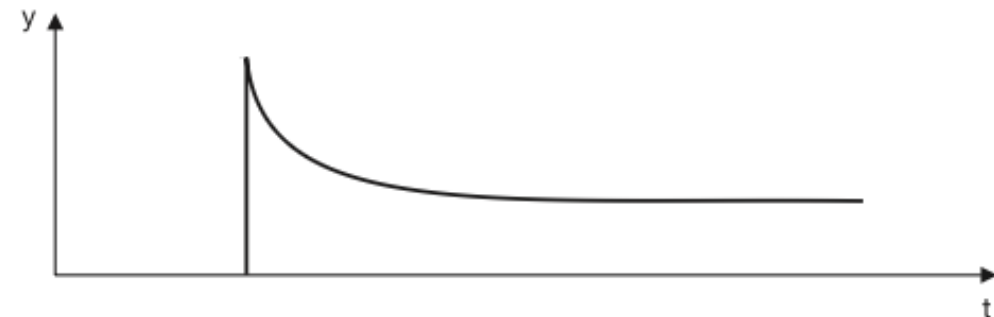
Step responses
P controller



PI controller



PD controller



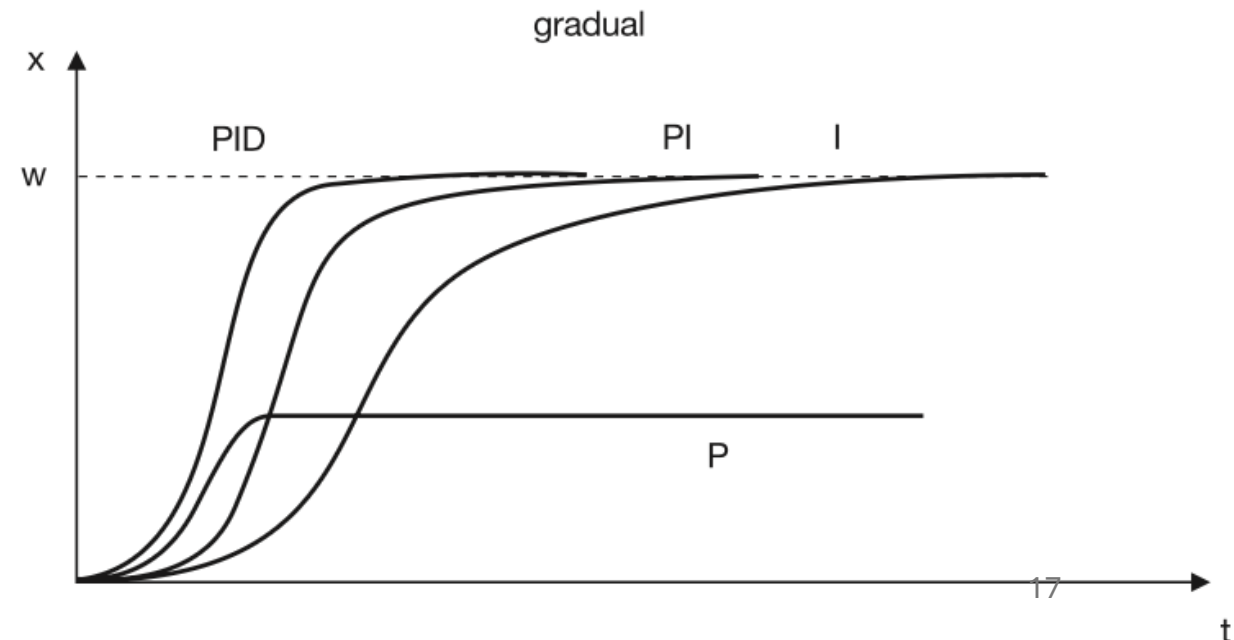
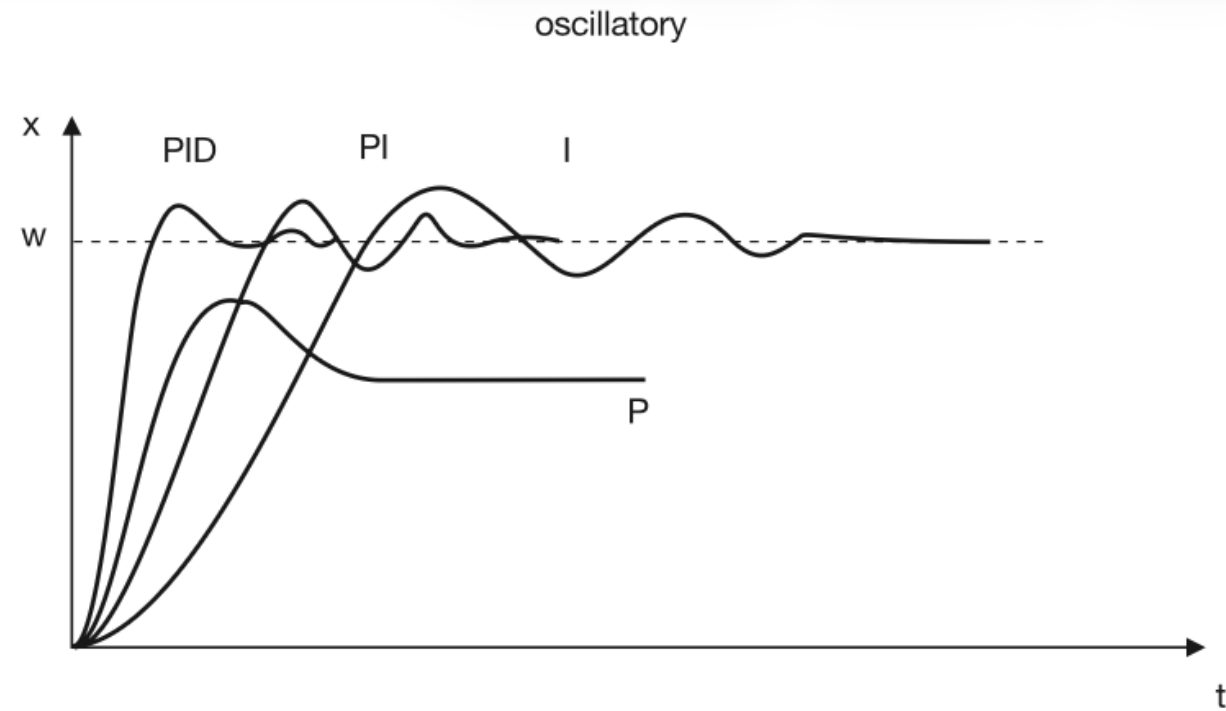
PID Controller

- Simple and intuitive
 - Works for many plants
 - Easy to tune experimentally
 - No model of process needed
- ~90% of industrial controllers are PID variants

P = speed

I = accuracy

D = damping



Time-Domain Performance Trade-Offs

Faster response vs. stability:

Higher $K_p \rightarrow$ faster response, but more overshoot and risk of instability

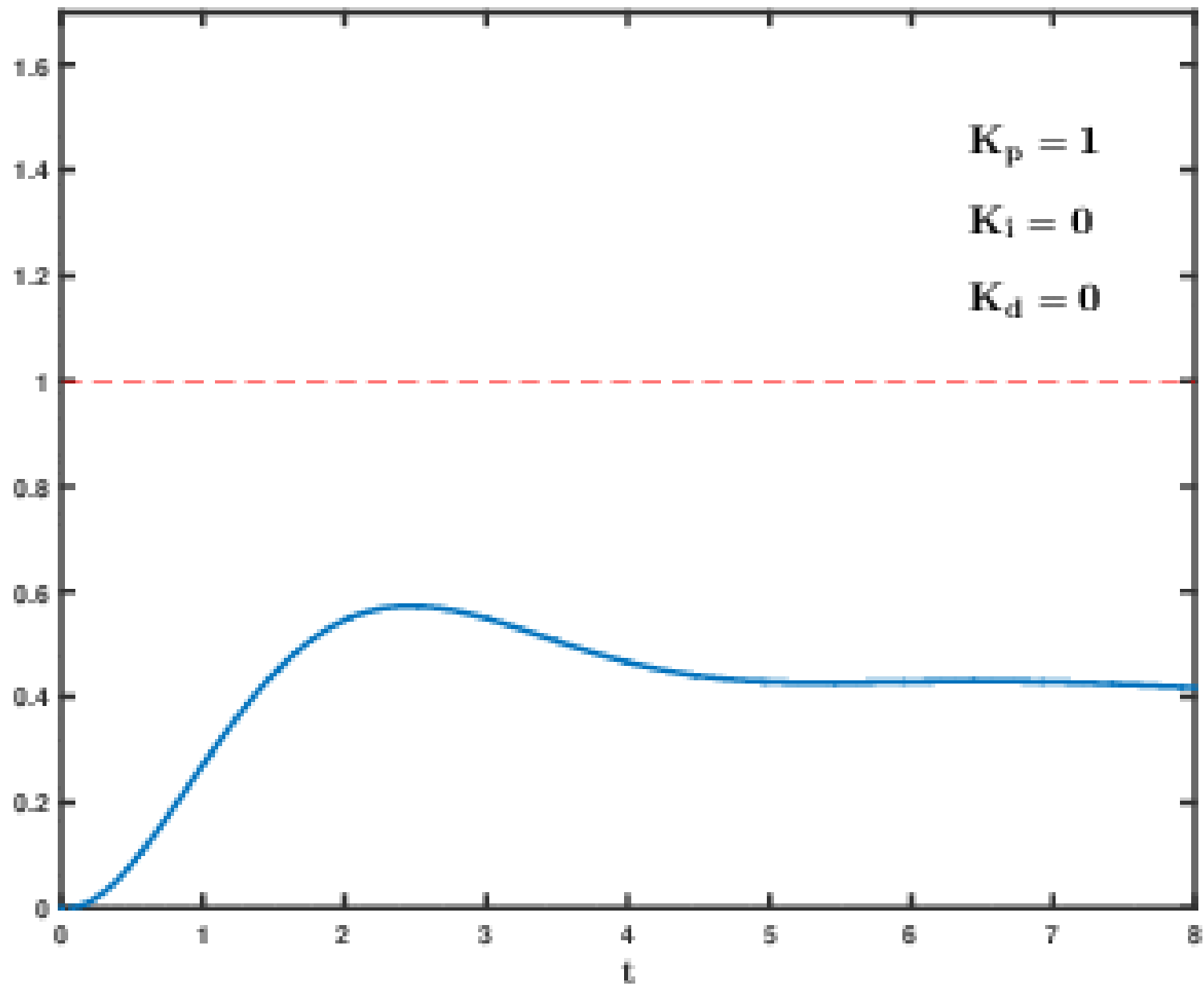
Accuracy vs. robustness:

Higher $K_i \rightarrow$ higher accuracy, but more sensitive to disturbances

Damping vs. noise sensitivity:

Higher $K_d \rightarrow$ less overshoot, but amplifies measurement noise

[Animated GIF](#)

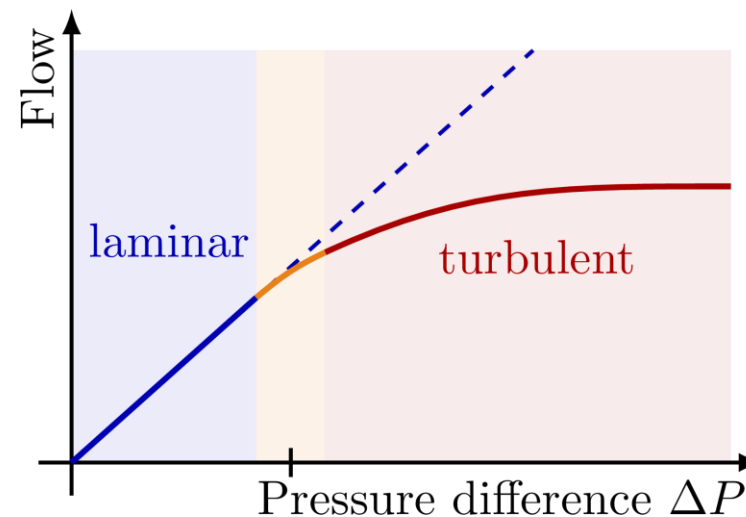
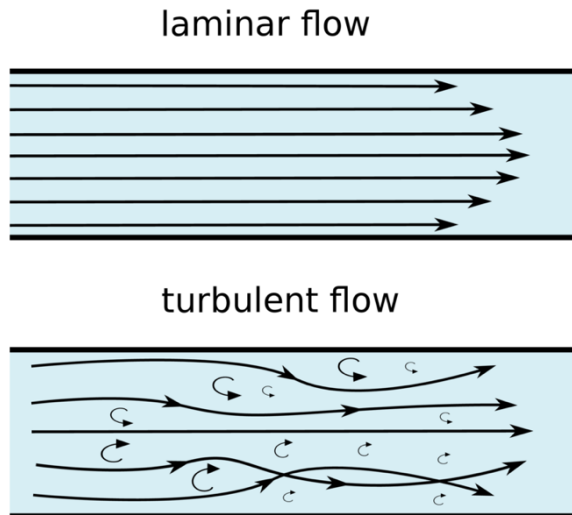


Nonlinear Systems

Linear models (output proportional to the input) are often used because they are easier to analyze.

However, real systems are typically nonlinear.

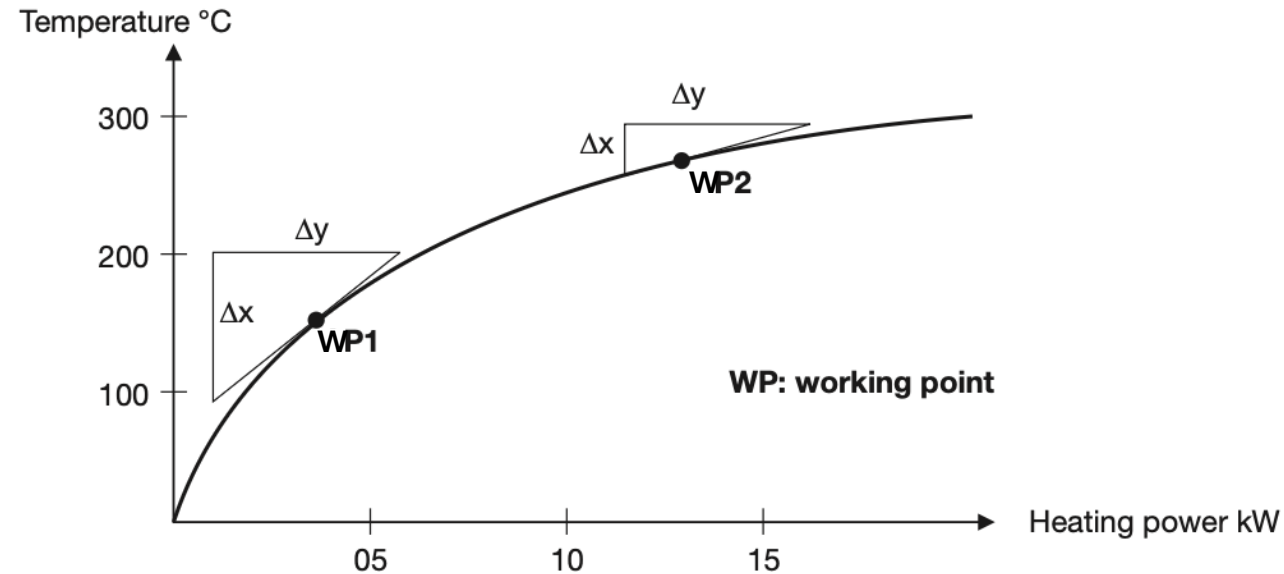
Example: flow resistance (laminar vs turbulent)



Working Point and Linearization

- Controllers designed and tuned **around** the operating/working point of the system
- Linear approximation of nonlinear systems (often near equilibrium)
- Valid locally

Example: furnace



⇒ K_S depends on the working point

$$\text{WP1: } K_S = \frac{\Delta x}{\Delta y} = \frac{100 \text{ °C}}{5 \text{ kW}} = 20 \frac{\text{°C}}{\text{kW}}$$

$$\text{WP2: } K_S = \frac{\Delta x}{\Delta y} = \frac{30 \text{ °C}}{5 \text{ kW}} = 6 \frac{\text{°C}}{\text{kW}}$$

Proportional Band

Range of error over which the controller moves its output (the input of the process to be controlled) from 0% to 100%

Inverse of proportional gain: $X_p = \frac{1}{K_p} \cdot 100\%$

Smaller band → stronger control action (from 0% to 100% over a narrower error range)

Common industrial specification

Exercise 1

Consider a first-order plant with $K = 1$, $T = 1$.

The system is controlled by a proportional controller with $K_p = 1$.

The input is a unit step setpoint.

- Determine the steady-state output and steady-state error.
- How do the results change for $K_p = 5$?
- How do the results change for $K = 2$?
- How do the results change for $T = 2$?
- How do the results change for a second-order plant with $K = 1$?

Exercise 2

Consider a first-order plant with $K = 1$, $T = 1$.

The system is controlled by a PI controller with $K_p = 1$ and $K_i = 1$.

The input is a unit step setpoint.

- Determine the steady-state output and steady-state error.
- How do the results change for $K_p = 5$?
- How do the results change for $K_i = 5$?
- How do the results change for $K = 2$?
- How does increasing K_i affect the transient response (qualitatively)?

Derivation for Exercise 1

- P controller: $u = K_p \cdot e$
 - error: $e = r - y$
 - steady-state output: $y = K \cdot u = K \cdot K_p \cdot e$
- steady-state error: $e = r - K \cdot K_p \cdot e$
- $e = \frac{r}{1 + K \cdot K_p}$